

Element G:

Construction Of A Testable Prototype

We created a dome which was more in a shape of an extremely flat cone, which allowed it to cover majority of the needed vision. We made sure it met all the requirements by scaling it based on our final design. The dome was formed by curling a circular piece of cardboard with reflective tape on one side and taping it down to create the cone shape. Unfortunately, the image was blurred due to the texture of the cardboard, the manner in which the cardboard was folded, and the manner in which the reflective side quickly picked up fingerprints.

To ensure it met all of the requirements, tests were run on the selected materials in order to ensure reflection and visibility. Even though this image did not have to be perfectly clear, it was important that the desired target was detectable.

We tested viability by having students stand at certain angles from the mirror and ask the librarian if she could see students in the mirror from her desk.

After debriefing with a college professor on our design idea to help the librarian issue, we received useful feedback which helped justify our design idea and further help ensure its effectiveness. Some key notes which the expert gave our team included mapping out the layout of the library beforehand on a precise scale to prevent errors, as well as finding a smoother backboard for the reflective paper to make the mirror more effective.

In order to complete the final design we will need to build a larger scale model, which will address the visibility issues present in the smaller prototype.



